SIGOPS

Infrastructure Execution OS — Open Source Core + Cloud

MIT License (Core) | Rust Agent | SEL Language | ~10 Tables | ~25 Endpoints

Version v1.2.0 | April 2026
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1. Executive Summary

SigOps is an infrastructure execution OS that automates DevOps and SRE workflows. Open source core (MIT license) + proprietary cloud. Features: signal ingestion, SEL (SigOps Execution Language), closed-loop execution engine, Rust-based lightweight agent, tool registry, workflow orchestration, and 6 marketplace plugins (SigShield, SigLens, SigVault, SigChaos, SigDeploy, SigCost). ~10 tables, ~25 endpoints. Monitors ALL ClusterAssets products.

Metric	Value
Entity	SigOps Technologies Pvt Ltd
Tables	~10
Endpoints	~25
Agent	Rust (~5MB binary, <20MB RAM)
License	MIT (core), Proprietary (cloud)

2. Product Overview

2.1 What It Is

SigOps receives signals from any infrastructure (alerts, metrics, logs, events), evaluates them against rules written in SEL, executes automated responses, and verifies the result — closed-loop.

2.2 What It Is NOT

Not a monitoring tool (Datadog). Not an alerting tool (PagerDuty). Not CI/CD (Jenkins). SigOps is the EXECUTION layer that sits after monitoring.

3. Core Modules

#	Module	Description
1	Signal Ingestion	Receives alerts from Prometheus, Grafana, Datadog, PagerDuty, CloudWatch, custom webhooks. Normalized into standard signal format.
2	SEL Engine	SigOps Execution Language — domain-specific language for infrastructure automation rules. Human-readable, version-controlled, testable.
3	Execution Engine	Closed-loop: Receive signal > Evaluate SEL > Execute actions > Verify result > Report. Retry with backoff, rollback on failure.
4	Agent Gateway	WebSocket connection to Rust agents on target infrastructure. Agent discovery, heartbeat, command dispatch.
5	Tool Registry	5 built-in tools (restart, scale, notify, http, script) + extensible custom tools. Agents discover available tools on boot.
6	Workflow Manager	Multi-step orchestration, parallel/sequential execution, conditional branching, approval gates. Schedule recurring workflows.
7	SigShield (Plugin)	Security automation: vulnerability scanning, compliance checks, auto-remediation.
8	SigLens (Plugin)	Log analysis: pattern detection, anomaly identification, root cause suggestion.
9	SigVault (Plugin)	Secret management: rotation, audit, integration with HashiCorp Vault.
10	SigChaos (Plugin)	Chaos engineering: fault injection, resilience testing.
11	SigDeploy (Plugin)	Deployment automation: blue/green, canary, rollback.
12	SigCost (Plugin)	Cloud cost optimization: unused resource detection, right-sizing recommendations.

4. Technology Stack

Unified ClusterAssets stack: Hono (backend), Drizzle ORM + PostgreSQL, Redis + BullMQ, TypeScript strict ESM, React 18 + Vite + Zustand + MUI, Zod, Pino, JWT HS256 via jose. All AI calls routed through UAP AI Management.

5. Complete Data Model

Table: signals

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
tenant_id	text	NOT NULL INDEX	
source	text	NOT NULL	e.g., prometheus, grafana
severity	enum	NOT NULL	critical/warning/info
title	text	NOT NULL	
body	jsonb	NOT NULL	Raw signal payload
fingerprint	text	NOT NULL	Dedup hash
status	enum	DEFAULT 'OPEN'	OPEN/ACKNOWLEDGED/RESOLVED
matched_workflow_id	text FK		
created_at	timestamp	DEFAULT now()	INDEX

Table: workflows

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
tenant_id	text	NOT NULL	
name	text	NOT NULL	
description	text		
sel_code	text	NOT NULL	SEL source code
trigger_rules	jsonb	NOT NULL	Signal matching rules
is_active	boolean	DEFAULT true	
version	integer	DEFAULT 1	
created_by	text FK		
created_at	timestamp	DEFAULT now()	
updated_at	timestamp	DEFAULT now()	

Table: executions

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
tenant_id	text	NOT NULL INDEX	
workflow_id	text FK	NOT NULL REFERENCES workflows	
signal_id	text FK		
status	enum	DEFAULT 'PENDING'	PENDING/RUNNING/SUCCESS/

			FAILED/ROLLED_BACK
trigger_type	text	NOT NULL	signal/manual/schedule
steps	jsonb	NOT NULL DEFAULT '[]'	Step results array
current_step	text		
error	jsonb		
started_at	timestamp		
completed_at	timestamp		
duration_ms	integer		
created_at	timestamp	DEFAULT now()	

Table: agents

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
tenant_id	text	NOT NULL	
hostname	text	NOT NULL	
ip_address	text		
os	text		
version	text	NOT NULL	Agent version
status	enum	DEFAULT 'ONLINE'	ONLINE/DEGRADED/OFFLINE
tools	jsonb	DEFAULT '[]'	Discovered tools
last_heartbeat	timestamp		
created_at	timestamp	DEFAULT now()	

Table: tool_registry

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
name	text	NOT NULL UNIQUE	Tool name
version	text	NOT NULL	
type	enum	NOT NULL	builtin/custom
schema	jsonb	NOT NULL	Input/output schema
description	text		
created_at	timestamp	DEFAULT now()	

Additional tables: signal_rules, execution_steps, execution_results, agent_tools, workflow_schedules.

6. Complete API Specification

Signals (4)

Method	Path	Auth	Description
POST	/api/v1/signals/ingest	Bearer/ webhook	Ingest signal. Body: { source, severity, title, body }
GET	/api/v1/signals	Bearer	List signals (paginated, filter by status/severity)
GET	/api/v1/signals/search	Bearer	Search signals by content
PUT	/api/v1/signals/:id/acknowledge	operator+	Acknowledge signal

SEL (4)

Method	Path	Auth	Description
POST	/api/v1/sel/validate	Bearer	Validate SEL code. Body: { code }
POST	/api/v1/sel/parse	Bearer	Parse SEL to AST
POST	/api/v1/sel/test	Bearer	Test SEL against sample signal
POST	/api/v1/sel/deploy	operator+	Deploy SEL to workflow

Executions (5)

Method	Path	Auth	Description
POST	/api/v1/executions/trigger	operator+	Manual trigger. Body: { workflow_id, params }
GET	/api/v1/executions	Bearer	List executions
GET	/api/v1/executions/:id	Bearer	Execution detail with steps
POST	/api/v1/executions/:id/cancel	operator+	Cancel running execution
POST	/api/v1/executions/:id/retry	operator+	Retry failed execution

Agents (4)

Method	Path	Auth	Description
GET	/api/v1/agents	Bearer	List agents
POST	/api/v1/agents/register	agent	Register new agent
POST	/api/v1/agents/:id/heartbeat	agent	Agent heartbeat + tool discovery
DELETE	/api/v1/agents/:id	operator+	Deregister agent

Tools & Workflows (8)

Method	Path	Auth	Description
GET	/api/v1/tools	Bearer	List tools
POST	/api/v1/tools	operator+	Register custom tool
POST	/api/v1/tools/:id/test	operator+	Test tool execution
GET	/api/v1/workflows	Bearer	List workflows

POST	/api/v1/workflows	developer+	Create workflow
PUT	/api/v1/workflows/:id	developer+	Update workflow
DELETE	/api/v1/workflows/:id	operator+	Delete workflow
POST	/api/v1/workflows/:id/schedule	operator+	Set schedule

7. Inter-Product Bonds

From	To	Service	Billing
SigOps	UAP	Auth, billing, AI, store	Platform fee
All products	SigOps	Infrastructure monitoring	SigOps subscription
SigOps	Armexa	Agent identity verification	Per-agent

8. RBAC — Roles Registered with UAP

Role	Permissions
sigops_admin	*
operator	signals:*, executions:*, workflows:*, tools:read
developer	signals:read, executions:read, workflows:rw, sel:rw
viewer	signals:read, executions:read (read-only)

9. Business Analysis

9.1 Target Market

DevOps/SRE teams globally ($12.8B market, 22% CAGR). MSPs, cloud consultancies, any company with 10+ servers. Open source community (GitHub-driven adoption).

9.2 Monetization

Stream	Pricing	Type
Community (OSS)	Free forever (MIT)	Free
Starter	$29/month (25 agents)	Recurring
Pro	$99/month (100 agents, AI)	Recurring
Business	$299/month (500 agents)	Recurring
Enterprise	$999+/month (unlimited, SSO)	Recurring
Marketplace	25% commission	Commission
Enterprise license	$4,999-19,999/year	Annual

Table: signal_rules

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
tenant_id	text	NOT NULL	
name	text	NOT NULL	Rule name
condition	jsonb	NOT NULL	{ source, severity_gte, title_match, body_match }
workflow_id	text FK	REFERENCES workflows	Auto-trigger workflow
dedup_window_sec	integer	DEFAULT 300	Dedup window
is_active	boolean	DEFAULT true	
created_at	timestamp	DEFAULT now()	

Table: execution_steps

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
execution_id	text FK	NOT NULL REFERENCES executions ON DELETE CASCADE	
step_index	integer	NOT NULL	0-based order
tool_name	text	NOT NULL	Tool to execute
input	jsonb	NOT NULL	Tool input params
output	jsonb		Tool output

status	enum	DEFAULT 'PENDING'	PENDING/RUNNING/SUCCESS/ FAILED/SKIPPED
agent_id	text FK		Target agent
started_at	timestamp		
completed_at	timestamp		
error	text		

Table: workflow_schedules

Column	Type	Constraints	Description
id	text PK	DEFAULT uuid()	
workflow_id	text FK	NOT NULL REFERENCES workflows	
tenant_id	text	NOT NULL	
cron_expression	text	NOT NULL	
timezone	text	DEFAULT 'UTC'	
is_active	boolean	DEFAULT true	
next_run_at	timestamp		
last_run_at	timestamp		
created_at	timestamp	DEFAULT now()	

9.3 Unit Economics

Per-user infra: Rs.2-5/mo. AI: Rs.10-30/mo. Avg revenue: $99/mo (Rs.8,250). Cost: Rs.200/mo. Gross margin: 97.6%. LTV:CAC ratio: 99x.

9.4 Financial Projections

Scenario	Revenue	Costs	Profit
Worst (Y1)	$1,450/mo (Rs.1.2L)	Rs.10K/mo	Rs.1.1L/mo
Realistic (Y2)	$49,500/mo (Rs.41L)	Rs.2L/mo	Rs.39L/mo
Best (Y3)	$500K+/mo	Rs.15L/mo	Rs.4Cr+/mo

9.5 Competitive Position

PagerDuty ($300M rev — alerting only), Datadog ($2.1B — monitoring only), Rundeck (OSS — execution only), StackStorm (OSS — complex). SigOps: closed-loop (signal > execute > verify), lightweight Rust agent, SEL language, AI-powered.

10. Security Model

Authentication via UAP (JWT HS256). RBAC per product. Tenant isolation on all queries. Audit logging via UAP. Encryption: TLS 1.3 transit, AES-256-GCM at rest.

11. MVP Scope

11.1 Must Launch

Signal ingestion (webhook + Prometheus + Grafana), SEL parser + evaluator, Basic execution engine, Agent WebSocket gateway, 5 built-in tools, Workflow CRUD + scheduler, Stats dashboard API.

11.2 Can Wait (V2)

AI-powered SEL generation, Advanced plugins (SigShield, SigChaos), Multi-region agent mesh, Marketplace, Chaos engineering, Cost optimization.